[Date]

The Editor-in-Chief
Agronomy

Dear Editor,

We submit the manuscript "Sensor-Driven Tilting Agrovoltaic Canopy for Sustainable Farming: Design, Simulation, and Prototype at TCE Madurai" for consideration as a research article in Agronomy.

Agrovoltaic systems share land between crops and solar panels, but a fixed panel angle favours either the crop or the electricity. We define a co-optimization accuracy that scores a tilt strategy against the best possible angle for the same objective, and we present a six-sensor tilt controller that reaches 99.6 to 99.7 percent of that best score at seven climates in simulation, against 89.1 to 91.5 percent for a fixed 45 degree tilt. The paper also reports a simulated month driven by real NASA POWER weather and describes a wire-driven canopy and a scaled prototype.

Why this journal (our judgement, please check the journal's aims and scope): Crop-focused journal that publishes agrivoltaic work; the crop light and water parts fit.

What we state openly: all results are simulation and the sensors are software models of named commercial devices; the accuracy is measured against the controller's own objective, not against crop yield; no field measurement is reported. A crop journal will weigh crop data heavily; this is the weakest scope fit because the crop model is not validated in the field.

[Authors to confirm before sending: the work is original, has not been published and is not under consideration elsewhere; all authors have approved the manuscript; there is no conflict of interest, or state it.]

Code, data and results are available at https://github.com/thanwin19/agrovoltaic-canopy-srmc.

Suggested reviewers: [to be completed by the authors].

Sincerely,
[corresponding author name, affiliation, e-mail]
Thanwin S S, Chandru P M, Kuberaeshwaran S, Mukilan A, and M. Meenakshi Devi
Thiagarajar College of Engineering, Madurai, India